

## Chomsky Hierarchy: Regular Grammars

Name: Abraham Ladha, John Hill

## 1.1 Properties

We said every regular language is context free. Any context free language can be decided by a pushdown automata. You can convert any finite automata to a pushdown automata by simply ignoring the stack. The context free languages are also characterized by context free grammars, so it follows that there should be a class of grammars which enumerate regular languages.

## 1.2 Regular Grammars

**Definition 1.1** A right-regular grammar is a tuple  $(V, \Sigma, P, S)$  such that:

- $V$  is a set of nonterminals, sometimes called variables. These are always denoted by upper case letters
- $\Sigma$  is the finite alphabet, denoted by lower case letters.
- $P$  is a set of production rules, only of the form:
  - $A \rightarrow aB$
  - $A \rightarrow a$
  - $A \rightarrow \epsilon$
- $S \in V$  is the starting symbol.

**Theorem 1.2** A language is regular if and only if there exists a regular grammar to generate it.

First we show for every right-regular<sup>1</sup> grammar, there exists an equivalent NFA for the same language.

Given a regular grammar  $G = (V, \Sigma, P, S)$ , we construct an NFA  $N = (\Sigma, Q, q_0, \delta, F)$

- $\Sigma$  is the same
- $Q$  is a set of states, one for each non-terminal, plus an additional state.  $Q = V \cup \{A\}$
- Recall that for an NFA,  $\delta(\cdot, \cdot)$  is a set. For each nonterminal  $B$  and character  $a$ :
  - For each nonterminal  $C$ , if  $B \rightarrow aC$ , then  $C \in \delta(B, a)$
  - If  $B \rightarrow a$ , then  $A \in \delta(B, a)$
  - For each  $a \in \Sigma$ , define  $\delta(A, a) = \emptyset$

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<sup>1</sup>left-regular grammars correspond to the reverse of the languages. It has rules of the form  $A \rightarrow Aa$ . This is not something we care about so much. The expressive power of regular languages, and their corresponding machines are invariant to this choice. We choose right-regular because of pedagogy.

- If  $B \rightarrow \epsilon$ ,  $A \in \delta(S, \epsilon)$
- $q_0$ , the start state is associated with the starting nonterminal,  $S$ .
- $F = \{A\}$ .

We show  $L(N) = L(G)$  by set containment

$(L(G) \subseteq L(N))$  Let  $x = a_1a_2\dots a_n \in L(G)$ . Then there exists a production of  $x$  of the form

$$S \Rightarrow a_1A_1 \Rightarrow a_1a_2A_2 \Rightarrow \dots \Rightarrow a_1a_2\dots a_{n-1}A_{n-1} \Rightarrow x \quad (1.1)$$

As defined, it is clear  $A_1 \in \delta(S, a_1)$ , and  $A_i \in \delta(A_{i-1}, a_i)$ . It follows then that  $A \in \delta(S, x)$ , and  $A \in F \Rightarrow x \in L(N)$ .

$(L(N) \subseteq L(G))$  Let  $x \in L(N)$ . If  $x = \epsilon$ , then we must have that  $A \in \delta(S, \epsilon)$ , but by construction this implies that  $S \rightarrow \epsilon$  is a rule of  $G$ . So suppose  $x \neq \epsilon$ . Then there exists a sequence of states,  $S, A_1, \dots, A_{n-1}, A$  such that  $A_1 \in \delta(S, a_1)$ , and  $A_i \in \delta(A_{i-1}, a_i)$ . This implies that we must have production rules of the form

$$S \Rightarrow a_1A_1 \Rightarrow a_1a_2A_2 \Rightarrow \dots \Rightarrow a_1a_2\dots a_{n-1}A_{n-1} \Rightarrow x \quad (1.2)$$

So  $G$  derives  $x$  and  $x \in L(G)$ .

Now we show for every NFA, there exists an equivalent right-regular grammar. Let  $D = (Q, \Sigma, \delta, q_0, F)$  be a DFA<sup>2</sup> We construct a grammar  $G = (V, \Sigma, P, S)$

- Let  $V$  correspond to the states of  $Q$
- $\Sigma$  is the same
- For each rule of the form  $\delta(B, a) = C$ , construct a production rule of the form  $B \rightarrow aC$ . If  $\delta(B, a) = C$ , and  $C \in F$ , then  $B \rightarrow a$ . If  $q_0 \in F$ , then add the production rule  $S \rightarrow \epsilon$
- $S$  corresponds to the start state,  $q_0$ .

The proof is left as exercise.

### 1.3 Problems

1. It is true, that a regular grammar will always generate a regular language, and for each regular language, there exists a regular grammar to produce it. But can a regular language be generated by a context free grammar which is not a regular grammar? If yes, give an example. If no, prove it.

**Answer.** Consider the following example,

$$\begin{aligned} S &\rightarrow aBa \\ B &\rightarrow b \end{aligned}$$

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<sup>2</sup>Not NFA, simply because the proof is easier. It doesn't matter, because you know DFAs and NFAs have the same expressive power.

Clearly this generates only  $aba$  and is trivially regular as it's finite. So for instance any context free grammar that is finite generates a regular language.

Another example is to consider the language  $a^m b^n$  where  $m, n \in \mathbb{N}$ . Here we have the grammar,

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow Aa \mid a \\ B &\rightarrow Bb \mid b \end{aligned}$$

This is a non-regular context free grammar that produces a regular language

2. We constructed a grammar  $G$  given an NFA  $M$ . Prove that  $S \xRightarrow{*} x \iff \delta(q_0, x) \in F$ . Just follow the reverse of the proof we showed.

**Answer.** So first we have  $S \xRightarrow{*} x$ , we need to show this implies that  $\delta(q_0, x) \in F$ . Clearly if  $S \xRightarrow{*} x$  then if  $x = a_1 \dots a_n$  then we have a series of productions of  $x$  of the form  $S \Rightarrow a_1 A_1 \Rightarrow a_1 a_2 \dots a_{n-1} A_{n-1} \Rightarrow x$ . Now as  $S$  corresponds to the start state  $q_0$  and for each non-terminal we have  $q_i$  corresponding to  $A_i$  (for  $i \in \{1, \dots, n-1\}$ ) note that if we traverse the string  $a_1, \dots, a_n$  we have  $q_i \in \delta(q_{i-1}, a_i)$  and if  $A_{n-1} \rightarrow a_n$  then we know have by definition  $q_n \in \delta(q_{n-1}, a_n)$  (for simplicity we have  $q_n = A$  instead i.e our accept state) so we have,

$$q_n \in \delta(\dots \delta(\delta(q_0, a_1), a_2), a_n) = \delta(q_0, x)$$

and we have  $q_n \in F$  by definition so we get  $\delta(q_0, x) \in F$

Now looking at the reverse we have  $\delta(q_0, x) \in F$  and we need to show that  $S \xRightarrow{*} x$ . Now if  $\delta(q_0, x) \in F$  then we have a sequence of state  $q_0, \dots, q_n$  such that  $q_n \in F$  and  $\delta(q_{i-1}, a_i) = q_i$ . Now clearly as  $q_0$  corresponds at  $S$  and we as each state  $q_i$  other than the accept and start state will correspond to some arbitrary non-terminal state say  $B_i$  we have if  $q_i \in \delta(q_{i-1}, a_i)$  this implies that  $B_{i-1} \rightarrow a_i B_{q_i}$  so putting it all together if we have  $\delta(q_0, x) = q_n \in F$  then this means we have,

$$\delta(q_0, x) = \delta(\dots \delta(\delta(q_0, a_1), a_2), a_n) \in F$$

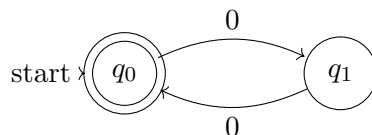
which means that we get,

$$S \Rightarrow a_1 A_1 \Rightarrow a_1 a_2 \dots a_{n-1} A_{n-1} \Rightarrow x$$

or  $S \xRightarrow{*} x$

3. Consider the regular language  $\{x \in \Sigma^* \mid x \text{ has an even number of 0s}\}$ . Give the minimal DFA for this language, and construct the right-regular grammar for that DFA (following the construction in the proof).

**Answer.** The minimal DFA is as follows,



Now in order to construct the right regular language first we have  $V = Q = \{q_0, q_1\}$  which is our set of non-terminals, let  $S = q_0$  and  $A = q_1$ . Now we're assuming our alphabet is  $\Sigma = \{0\}$  so we have two rules  $\delta(q_0, 0) = q_1$  and  $\delta(q_1, 0) = q_0$  which corresponds to the rules  $S \rightarrow 0A$  and  $A \rightarrow 0S$ . And as we have  $\delta(q_1, 0) = q_0$  and  $q_0 \in F$  we have  $q_1 = A \rightarrow 0$ . And as  $q_0 \in F$  we have  $S \rightarrow \epsilon$ . Putting this together we get,

$$\begin{aligned} S &\rightarrow 0A \mid \epsilon \\ A &\rightarrow 0S \mid 0 \end{aligned}$$

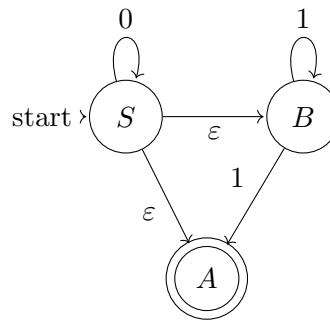
(If we assume we have other alphabets as well then in our DFA we just have all the symbols at  $q_0$  and  $q_1$  except 0 go to itself which just corresponds  $\delta(q_1, a) = q_1$  which is the production  $A \rightarrow aA$  for  $a \neq 0$  and  $\delta(q_0, a) = q_0$  which is  $S \rightarrow aS$  for  $a \neq 0$  and as  $S$  is the end state we also have  $S \rightarrow a$  for  $a \neq 0$ )

4. For the language  $\{0^i 1^j \mid i, j \geq 0\}$  give the minimal<sup>3</sup> right-regular grammar for this language, then construct the NFA associated with that grammar.

**Answer.** The minimal right regular grammar is,

$$\begin{aligned} S &\rightarrow 0S \mid B \mid \epsilon \\ B &\rightarrow 1B \mid \epsilon \end{aligned}$$

Now to convert it into an NFA we have,



5. Let  $A_1, A_2$  be any non terminals and let  $a_1, \dots, a_n$  be any terminals. Prove that a production rule of the form  $A_1 \rightarrow a_1 \dots a_n A_2$  can be converted to a set of production rules of a regular grammar.

**Answer.** What we need to show that there is an equivalent regular grammar to the above. We do this by introducing new non-terminals. Let,

$$B_1 \rightarrow a_2 \dots a_n A_2$$

then we have,

$$A_1 \rightarrow a_1 B_1$$

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<sup>3</sup>We haven't talked about what a minimal regular grammar is, so just give the smallest (with respect to number of nonterminals) one you can come up with

Now this leaves us with a non regular grammar for  $B_1$  which is,

$$B_1 \rightarrow a_2 \dots a_n A_2$$

but note that we have one less terminal in the left side. So inductively we do this  $n - 1$  times until we have,

$$B_{n-1} \rightarrow a_n A_2$$

So now we have the group of productions,

$$\begin{aligned} A_1 &\rightarrow a_1 B_1 \\ B_1 &\rightarrow a_2 B_2 \\ &\dots \\ B_{n-2} &\rightarrow a_{n-1} B_{n-1} \\ B_{n-1} &\rightarrow a_n A_2 \end{aligned}$$

Now note that each of this productions correspond to a regular grammar. So we're done.